MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
Model Question Paper
MG3DSECHE201 – INTRODUCTION TO NANOSCIENCE
(2024 ADMISSION ONWARDS)

Duration: 2 hrs Max. Marks: 70

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **5** questions. Each question carries **4** marks

1. What are semiconductor quantum dots? Give one method for its synthesis with example. [R] [CO: 3]
2. Compare the properties of single walled and multiwalled carbon nanotubes. [R] [CO:3]
3. List any four applications of nanomaterials in medicine and health care. [U] [CO:4]
4. What are piezo electric nanomaterials? How is it useful in energy storage? [U] [CO: 4]
5. Discuss the applications of scanning electron microscopy. [U] [CO:2]
6. Write short note on SIMS. [R] [CO: 2]
7. Classify nanomaterials based on their dimensions with examples. [U] [CO: 1]

Part B

Short Essay Type questions

Answer any 5 questions. Each question carries 7 marks

8. How does nature inspire nanotechnology? Demonstrate with examples. [A] [CO:1]
9. Discuss the advantages and applications of STM and Auger electron microscopy in nanoscience. [U] [CO: 2]
10. Explain the sensing applications of nanomaterials. Discuss how is it useful in defence purposes with suitable examples. [A] [CO:4]
11. Give one synthesis method for magnetic nanoparticles. Explain its properties and real-world applications. [A] [CO: 3]
12. Discuss the instrumental features, principle and applications of AFM. [U] [CO: 2]

13. How nanotechnology is useful in water purification and air pollution control? [U][CO:4]

14. Explain how IR, UV-visible and X-ray Photoelectron spectroscopy is helpful in the characterization of nanomaterials. [A][CO:2]

Part C

Essay Type questions

Answer any 1 question. Each question carries 15 marks

15. Illustrate the synthesis methods for the preparation of nanomaterials [U] [CO: 1]

16. A) Compare Scanning electron microscope and Transmission electron microscope and explain how it is helpful in nanoscience. B) Applications of nanotechnology in data storage and displays. [U] [CO:2,4]